

# Venkata Dhiraj Reddy Uppada

AI / Software Engineer · B.Tech Computer Science, 2027

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## SUMMARY

Computer Science undergraduate with hands-on experience building AI/ML and LLM-based applications in Python. Interested in AI engineering, software development and data/ML roles focused on building practical applications: agents that cite evidence, retrieval-based assistants, computer vision and backend systems.

## EDUCATION

|                                                                                                     |             |
|-----------------------------------------------------------------------------------------------------|-------------|
| <b>B.Tech, Computer Science &amp; Engineering</b> — Vidya Jyothi Institute of Technology, Hyderabad | 2023 – 2027 |
| <b>Intermediate (MPC)</b> — Resonance Junior College, Hyderabad                                     | 2021 – 2023 |

## SKILLS

**Languages:** Python, TypeScript/JavaScript, Java, SQL, HTML/CSS, C

**AI / ML:** Machine learning, NLP, RAG, AI agents, computer vision, scikit-learn, LangChain, Ollama, OpenCV

**Backend:** FastAPI, Django, SQLAlchemy, REST API design

**Frontend:** React, Next.js, Tailwind CSS, Streamlit

**Data:** PostgreSQL, MySQL, SQLite, ChromaDB, pandas, NumPy

**Tools:** Git & GitHub, Docker, Playwright, pytest

## PROJECTS

### Verascope — AI repository intelligence & debugging platform

FastAPI · PostgreSQL · ChromaDB · React · Docker · Ollama · AI-assisted (vibe-coded with Claude)

- Answers questions about a codebase with cited snippets; a debugging agent must report an evidence status or “insufficient evidence”.
- Fixes are tested in an isolated Docker sandbox and need human approval before any branch or PR; docs/VERIFICATION.md logs four test passes against real repositories.

### ScoutLens — opportunity-investigation agent

Python · Streamlit · Pydantic · SerpApi · SQLite · SerpApi India Hackathon 2026 · built with Claude Code

- Investigates a job or internship with Google Search, Jobs and News results; every finding must cite retrieved evidence (“NO SOURCE = NO FACT”).
- 592 automated tests pass without network access; live-checked against two real listings.

### SCARFLOW — supplier corrective-action (SCAR / 8D) workflow

Next.js 16 · TypeScript · Supabase (Postgres + RLS) · Inngest · Playwright · built with Claude Code

- Multi-tenant SaaS: hashed magic-link supplier portal without accounts, audit rules enforced by database triggers, daily deadline sweep.
- Playwright suite covering the full workflow on desktop and mobile plus direct-API security attacks.

### SmartRide AI — two-wheeler rider-safety assistant (team of four)

Python · Raspberry Pi · YOLOv8 · MediaPipe · OpenCV · MPU6050

- Implemented YOLOv8 object/helmet detection and MediaPipe hand tracking for real-time monitoring.
- Integrated MPU6050 motion sensing to trigger safety alerts through the camera-sensor pipeline.

### Fake News Predictor

Python · scikit-learn · TF-IDF · Streamlit

- Text classifier (TF-IDF + logistic regression) trained on an Indian fake-news dataset of 3,729 labelled articles.
- Streamlit app that downloads an article from its URL and returns a FAKE/REAL label with a confidence score.

## OPEN SOURCE

GirlScript Summer of Code 2026 — selected as a contributor for the AI Agent Track and Open Source Track.

